

$G \cdot H^T = 0$ correction

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$$G \cdot H^T = [I_{11} | P] \cdot [P^T | I_4]^T$$

$$= [I_{11} | P] \cdot \begin{bmatrix} P \\ I_4 \end{bmatrix} = I_{11} \cdot P + P \cdot I_4 = P \oplus P = 0$$

$$= \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}}_G \cdot \underbrace{\begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}}_{H^T}$$

$$g_0 \cdot h_0 = 1 \oplus 0 \oplus 0 \oplus 0 \oplus 0 \oplus 0 \oplus 0 \oplus 0 \oplus 0 \oplus 0 \oplus 0 \oplus 0 \oplus 1$$

$$\oplus 0 \oplus 0 \oplus 0 \oplus 0$$

$$= 1 \oplus 1 = 0$$

$G.H^T = 0$ correction

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$$G \cdot H^T = [I_{11} | P] \cdot [P^T | I_4]^T$$

$$= [I_{11} | P] \cdot \begin{bmatrix} P \\ I_4 \end{bmatrix} = I_{11} \cdot P + P \cdot I_4 = P \oplus P = 0$$

[illegible]

$$g_0 = [1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 1 \ 0]$$

$$H_0 = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \end{bmatrix}$$

$$g_0 \cdot H_0 = 1 \oplus 1 = 0$$

$$g_0 = [1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 1 \ 0]$$

$$H_1 = [0 \ 0 \ 1 \ 1 \ 1 \ 1 \ 0 \ 1 \ 1 \ 1 \ 0 \ 0 \ 1 \ 0 \ 0]$$

$$g_0 \cdot H_1 = 0$$

* All products are 0, so dot product result 0.

XOR table

a	b	Y
0	0	0
0	1	1
1	0	1
1	1	0

②

$$g_0 = [\begin{pmatrix} 1 \end{pmatrix} 0 0 0 0 0 0 0 0 0 0 1 0 \begin{pmatrix} 1 \end{pmatrix} 0]$$

$$H_2 = [\begin{pmatrix} 1 \end{pmatrix} 1 0 0 1 1 1 0 1 1 0 0 0 \begin{pmatrix} 1 \end{pmatrix} 0]$$

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$$g_0 \cdot H_2 = 1 \oplus 1 = 0$$

$$g_0 = [1 0 0 0 0 0 0 0 0 0 0 1 0 1 0]$$

$$H_3 = [0 1 0 1 0 1 1 1 0 1 1 0 0 0 1]$$

$$g_0 H_3 = 0$$

* All products are 0, so dot product result 0.

$$g_0 \cdot H^T = [0 0 0 0] \rightarrow \text{first row}$$

$$g_7 = [0 0 0 0 0 0 0 1 0 0 0 0 0 1 0 1]$$

$$H_0 = [1 1 1 1 1 1 0 0 0 0 1 1 0 0 0]$$

$$g_7 \cdot H_0 = 0$$

* All product are 0, so dot product result 0.

$$g_7 = [0 0 0 0 0 0 0 \begin{pmatrix} 1 \end{pmatrix} 0 0 0 0 \begin{pmatrix} 1 \end{pmatrix} 0 1]$$

$$H_1 = [0 0 1 1 1 1 0 \begin{pmatrix} 1 \end{pmatrix} 1 1 0 0 \begin{pmatrix} 1 \end{pmatrix} 0 0]$$

$$g_7 \cdot H_1 = 1 \oplus 1 = 0$$

$$g_7 = [0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 1]$$

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$$H_2 = [1 \ 1 \ 0 \ 0 \ 1 \ 1 \ 1 \ 0 \ 1 \ 1 \ 0 \ 0 \ 0 \ 1 \ 0]$$

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$$g_7 \cdot H_2 = 0$$

* All product are 0, so dot product result 0.

$$g_7 = [0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 1]$$

$$H_3 = [0 \ 1 \ 0 \ 1 \ 0 \ 1 \ 1 \ 1 \ 1 \ 0 \ 1 \ 1 \ 0 \ 0 \ 0 \ 1]$$

$$g_7 \cdot H_3 = 1 \oplus 1 = 0$$

$$g_7 \cdot H^T = [0 \ 0 \ 0 \ 0]$$

$$G \cdot H^T = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

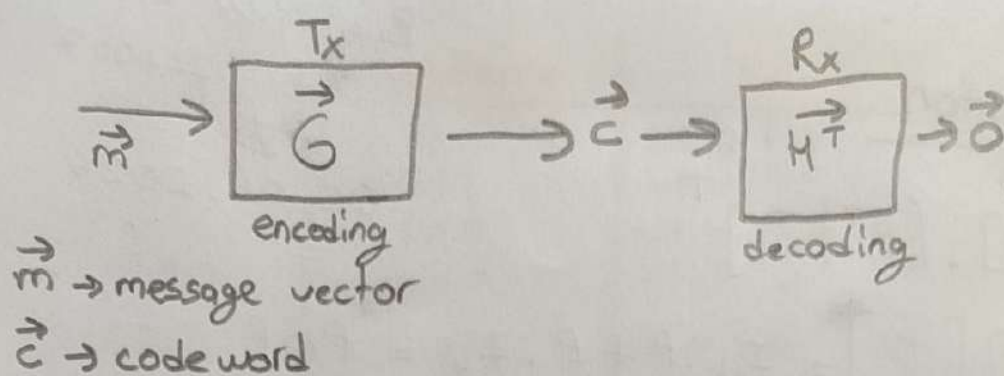
→ $g_0 \cdot H^T$

→ $g_7 \cdot H^T$

$$\text{So } G \cdot H^T = 0$$

$$C = mG$$

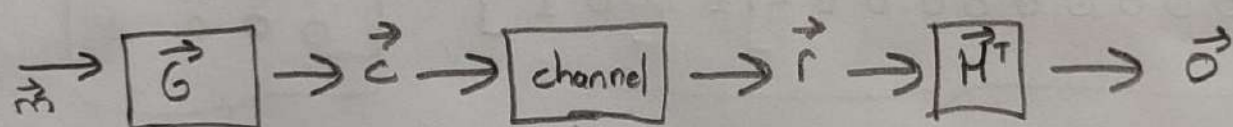
$$C \cdot H^T = m \underbrace{G \cdot H^T}_0 = m \cdot 0 = 0$$



★ Meanwhile, we systematically derived G from the matrix H .
 We verified that $G \cdot H^T = 0$ both algebraically and digitally.
 I also showed the system as block diagram.

Syndrome decoding

$\vec{r}$: $1 \times n$ received vector that results from sending the code vector $\vec{c}$ over a noisy channel.



$$\vec{r} = \vec{c} + \vec{e}$$

(Tx + channel)

$\vec{e}$ = error vector

$$e_i = \begin{cases} 1, & \text{if error occurred in the } i^{\text{th}} \text{ location} \\ 0, & \end{cases}$$

$$\vec{s} = \vec{r} H^T = (\vec{c} + \vec{e}) H^T = \underbrace{\vec{c} H^T}_0 + \vec{e} H^T$$

$$\vec{s} = \vec{e} H^T$$

IF $S=0 \rightarrow$ No error

$S \neq 0 \rightarrow$ Error

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Bit location	Syndrome	
0 (m_0)	1 0 1 0	bit 0 Flip
1 (m_1)	1 0 1 1	bit 1 Flip
2 (m_2)	1 1 0 0	bit 2 Flip
3 (m_3)	1 1 0 1	bit 3 Flip
4 (m_4)	1 1 1 0	bit 4 Flip
5 (m_5)	1 1 1 1	bit 5 Flip
6 (m_6)	0 0 1 1	bit 6 Flip
7 (m_7)	0 1 0 1	bit 7 Flip
8 (m_8)	0 1 1 0	bit 8 Flip
9 (m_9)	0 1 1 1	bit 9 Flip
10 (m_{10})	1 0 0 1	bit 10 Flip
11 (p_0)	1 0 0 0	bit 11 Flip
12 (p_1)	0 1 0 0	bit 12 Flip
13 (p_2)	0 0 1 0	bit 13 Flip
14 (p_3)	0 0 0 1	bit 14 Flip

0000 $\rightarrow$ no error

error analysis

$h_i \neq h_j$ so $h_i \oplus h_j = 0 \rightarrow$ linear independence

$$d_{\min} = 3 \rightarrow h_2 \oplus h_3 \oplus h_4 = 0$$

$d_{\min} \geq 2t + 1$ $t \rightarrow$ error correction capability

$$t \leq \frac{d_{\min} - 1}{2}$$

$t \leq 1$ ~~Hamming~~ Hamming can correct one-bit error.

~~Hamming~~ Hamming can detect two bit error. But can't correct it.

Conv. Code Parameters

rate $\Rightarrow 1/2$

$K=4$ (constraint length)

$M=K-1=3$ $2^3=8$ state

$g_1=1101 \rightarrow 15$ octal

$$g_1(D) = D^3 + D^2 + 1$$

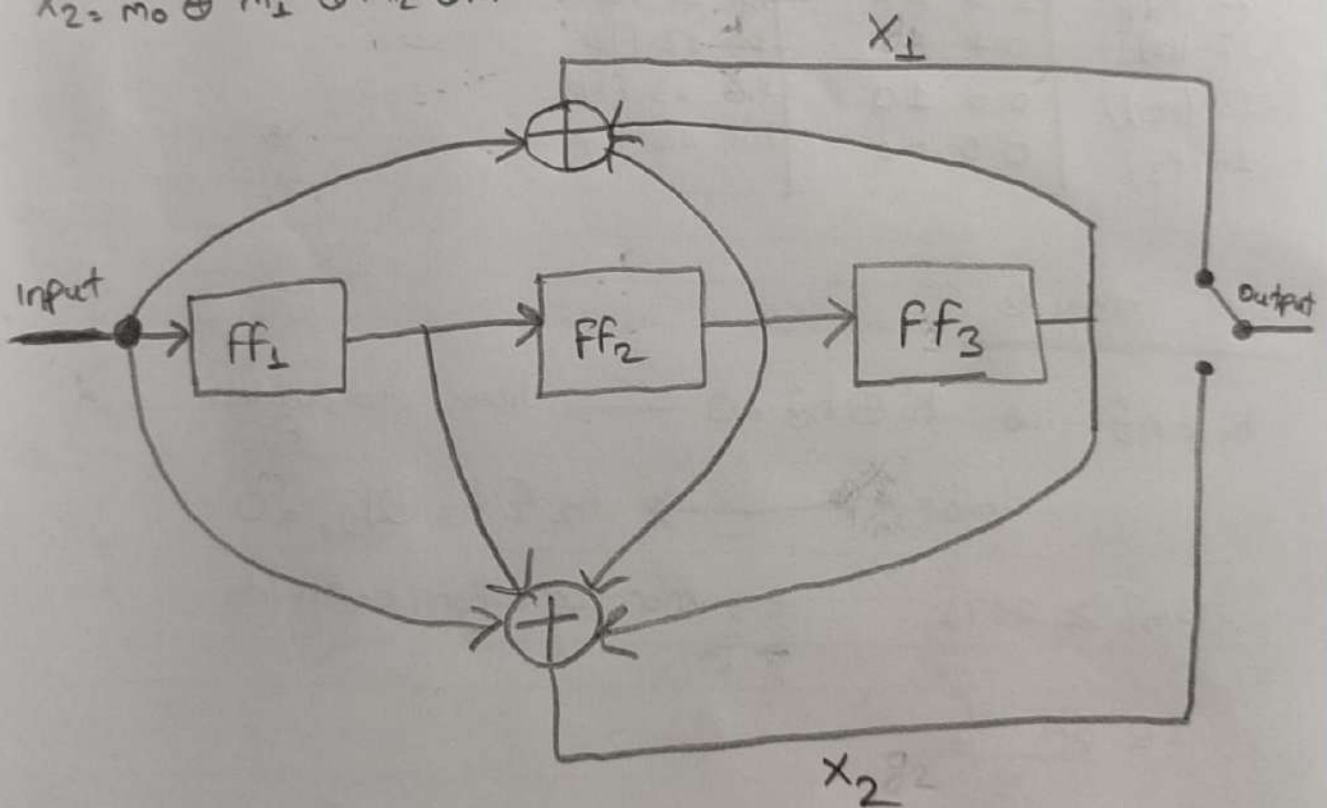
$g_2=1111 \rightarrow 17$ octal

$$g_2(D) = D^3 + D^2 + D + 1$$

$[m_0 \ m_1 \ m_2] \rightarrow m_0$: oldest bit m_1 : added bit

$$X_1 = m_0 \oplus m_1 \oplus m$$

$$X_2 = m_0 \oplus m_1 \oplus m_2 \oplus m$$



$$X_1 = m_0 \oplus m_1 \oplus m$$

$$X_2 = m_0 \oplus m_1 \oplus m_2 \oplus m$$

$n \cdot (m \cdot k) \rightarrow$ output bits

$\downarrow \quad \downarrow \quad \downarrow$
2 3 4

14 bit

$\underbrace{X_1 X_2 \ X_1 X_2 \ X_1 X_2 \ X_1 X_2 \ X_1 X_2 \ X_1 X_2 \ X_1 X_2}_{\text{output}}$

(7)

$m = (1, 0, 1, 1) \rightarrow$ uncoded message

$$m(D) = 1 + D^2 + D^3 \quad 0 \rightarrow 3$$

$$g_1(D) = D^3 + D^2 + 1 \quad 3 \rightarrow 0$$

$$g_2(D) = D^3 + D^2 + D + 1$$

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Path-1

$$x_1 = c_1(D) = g_1(D) \cdot m(D)$$

$$= (D^3 + D^2 + 1) \cdot (1 + D^2 + D^3)$$

$$= D^3 + D^2 + 1 + D^5 + D^4 + D^2 + D^6 + D^5 + D^3$$

$$= D^6 + \underbrace{2D^5}_0 + D^4 + \underbrace{2D^3}_0 + \underbrace{2D^2}_0 + 1$$

$$D^2 \oplus D^2 = 0$$

$$D^3 \oplus D^3 = 0$$

$$D^5 \oplus D^5 = 0$$

$$x_1 = c_1(D) = D^6 + D^4 + 1$$

$$[1 \ 0 \ 0 \ 0 \ 1 \ 0 \ 1] \quad D^0 \rightarrow D^6$$

$$x_2 = c_2(D) = g_2(D) \cdot m(D)$$

$$= (D^3 + D^2 + D + 1) \cdot (1 + D^2 + D^3)$$

$$= D^3 + D^2 + D + 1 + D^5 + D^4 + D^3 + D^2 + D^6 + D^5 + D^4 + D^3$$

$$= D^6 + \underbrace{2D^5}_0 + \underbrace{2D^4}_0 + \underbrace{3D^3}_0 + \underbrace{2D^2}_0 + D + 1$$

$$x_2 = c_2(D) = D^6 + D^3 + D + 1$$

$$[1 \ 1 \ 0 \ 1 \ 0 \ 0 \ 1] \quad D^0 \rightarrow D^6$$

Path 1 and Path 2
mux

11 01 00 01 10 00 11

8

State diagram

$K=4$, so $2^3 = 8$ state
 $M=K-1=3$ $2^M \rightarrow$

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m_0	m_1	m_2	current state
0	0	0	a
0	0	1	b
0	1	0	c
0	1	1	d
1	0	0	e
1	0	1	f
1	1	0	g
1	1	1	h

$$X_1 = m_0 \oplus m_1 \oplus m$$

$$X_2 = m_0 \oplus m_1 \oplus m_2 \oplus m$$

m_0	m_1	m_2	m	X_1	X_2	current state	next state
0	0	0	0	0	0	a	b
0	0	0	1	1	1	a	b
0	0	1	0	0	1	b	c
0	0	1	1	1	0	b	c
0	1	0	0	1	0	c	d
0	1	0	1	0	1	c	d
0	1	1	0	0	1	d	e
0	1	1	1	1	0	d	e
1	0	0	0	1	1	e	f
1	0	0	1	0	0	e	f
1	0	1	0	0	1	f	g
1	0	1	1	1	0	f	g
1	1	0	0	1	0	g	h
1	1	0	1	0	1	g	h
1	1	1	0	0	1	h	a
1	1	1	1	1	0	h	a

current state = (m_0, m_1, m_2)

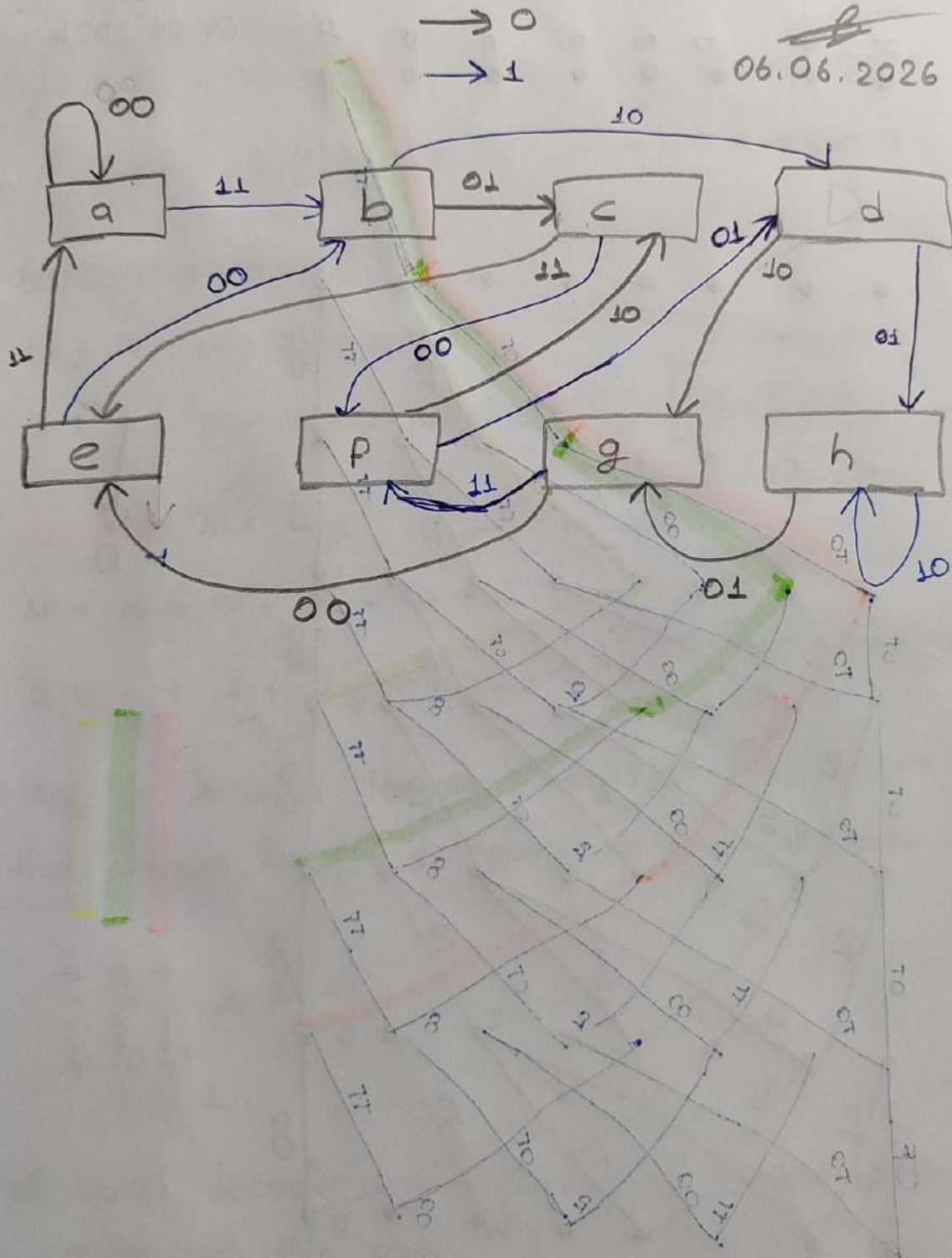
next state = (m_1, m_2, m)

State diagram

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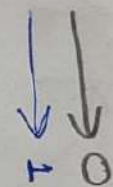
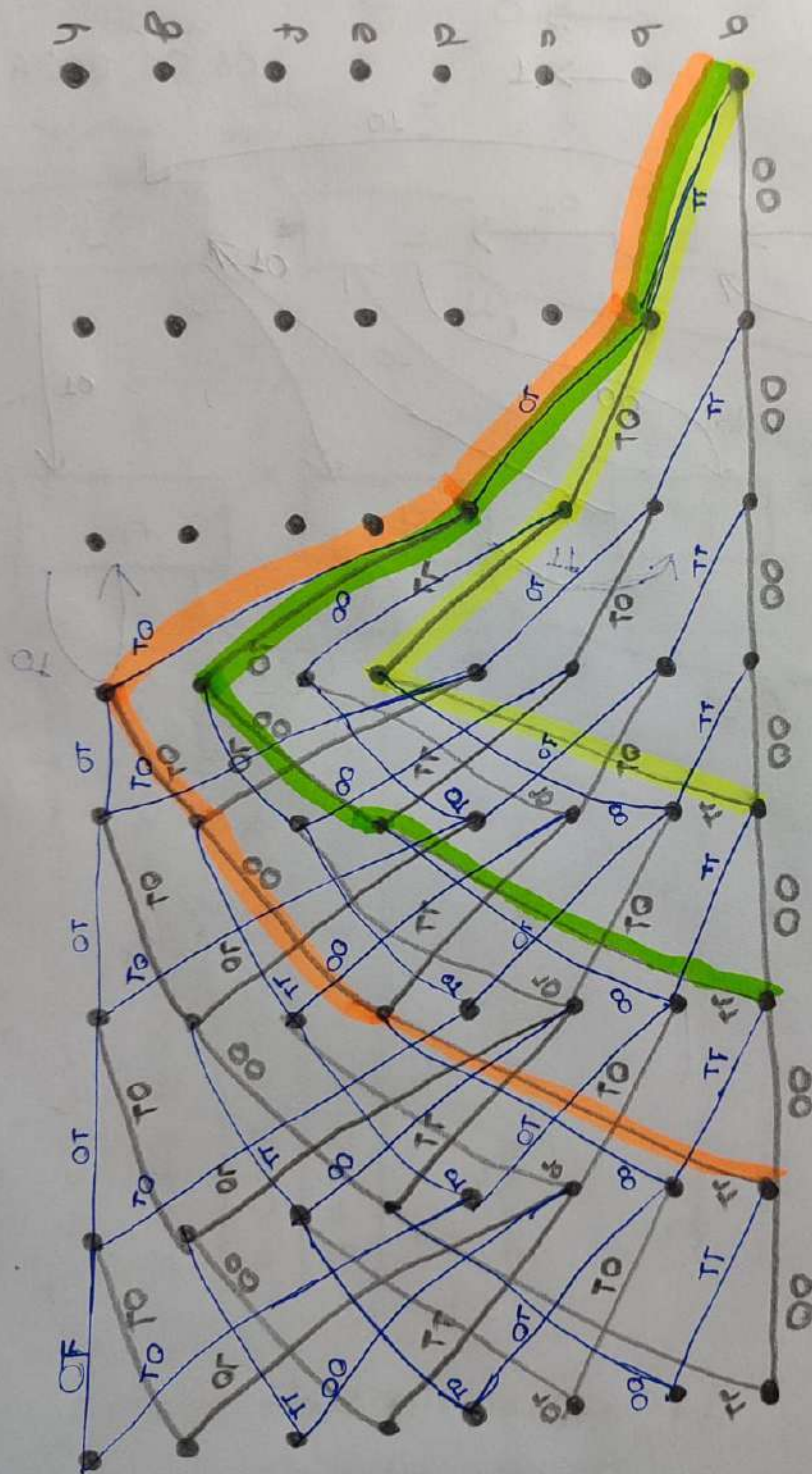
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TF

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First way

$$11 + 01 + 11 + 11$$

$$2 + 1 + 2 + 2 = 7 \quad \times$$

Second way

$$11 + 10 + 10 + 00 + 11$$

$$2 + 1 + 1 + 0 + 2 = 6 \quad \checkmark$$

$$d_{\text{free}} = 6$$

Third way

$$11 + 10 + 01 + 01 + 00 + 11$$

$$2 + 1 + 1 + 1 + 0 + 2 = 7 \quad \times$$

* d_{free} is minimum Hamming weight of any non-zero path that diverges from and returns to the all zeros state.

$$d_{\text{free}} \geq 2t + 1$$

$$t \leq 2$$

* Viterbi decoder can theoretically correct up to 2 bit errors with this code.

$$m = (1, 0, 1, 1)$$

↓ add 3 flush bits

$$(1, 0, 1, 1, 0, 0, 0)$$

$$\text{initial state} \rightarrow [m_0 \ m_1 \ m_2] = [0, 0, 0] \rightarrow a$$

$$\Rightarrow m = 1, \text{ register} = [0, 0, 0]$$

$$x_1 = m_0 \oplus m_1 \oplus m = 1$$

$$x_2 = m_0 \oplus m_1 \oplus m_2 \oplus m = 1$$

$$\begin{matrix} x_1 & x_2 \\ \text{output:} & 1 & 1 \end{matrix}$$

$$\text{current register: } [0, 0, 1] \rightarrow b$$

$$\Rightarrow m = 0, \text{ register} = [0, 0, 1],$$

$$x_1 = m_0 \oplus m_1 \oplus m = 0$$

$$x_2 = m_0 \oplus m_1 \oplus m_2 \oplus m = 1$$

$$\begin{matrix} x_1 & x_2 \\ \text{output} = & 0 & 1 \end{matrix}$$

$$\text{current register: } [0, 1, 0] \rightarrow c$$

$$\Rightarrow m = 1, \text{ register} = [0, 1, 0]$$

$$x_1 = m_0 \oplus m_1 \oplus m = 0$$

$$x_2 = m_0 \oplus m_1 \oplus m_2 \oplus m = 0$$

$$\begin{matrix} x_1 & x_2 \\ \text{output} = & 0 & 0 \end{matrix}$$

$$\text{current register: } [1, 0, 1] \rightarrow f$$

$$\Rightarrow m = 1, \text{ register} = [1, 0, 1]$$

$$x_1 = m_0 \oplus m_1 \oplus m = 0$$

$$x_2 = m_0 \oplus m_1 \oplus m_2 \oplus m = 1$$

$$\begin{matrix} x_1 & x_2 \\ \text{output} = & 0 & 1 \end{matrix}$$

$$\text{current register: } [0, 1, 1] \rightarrow d$$

$$\Rightarrow m = 0, \text{ register} = [0, 1, 1]$$

$$x_1 = m_0 \oplus m_1 \oplus m = 1$$

$$x_2 = m_0 \oplus m_1 \oplus m_2 \oplus m = 0$$

$$\begin{matrix} x_1 & x_2 \\ \text{output:} & 1 & 0 \end{matrix}$$

$$\text{current register: } [1, 1, 0] \rightarrow g$$

$\rightarrow m=0$, register = $[1, 1, 0]$

$$x_1 = m_0 \oplus m_1 \oplus m = 0$$

$$x_2 = m_0 \oplus m_1 \oplus m_2 \oplus m = 0$$

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~~B~~

$x_1 x_2$ 06.06.2026

output 00

current state: $[1, 0, 0] \rightarrow e$

$\rightarrow m=0$, register: $[1, 0, 0]$

$$x_1 = m_0 \oplus m_1 \oplus m = 1$$

$$x_2 = m_0 \oplus m_1 \oplus m_2 \oplus m = 1$$

$x_1 x_2$

output 11

current state: $[0, 0, 0] \rightarrow a$

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B

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m_0	m_1	m_2	m	x_1	x_2	current state	next state
0	0	0	1	1	1	a	b
0	0	1	0	0	1	b	c
0	1	0	1	0	0	c	r
1	0	1	1	0	1	f	d
0	1	1	0	1	0	d	g
1	1	0	0	0	0	g	e
1	0	0	0	1	1	e	a

11 01 00 01 10 00 11